

2Market Customer and Campaign Analysis

A portfolio case study using Excel, PostgreSQL, and Tableau

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Analysis reference year	2025
Customer records	2,216
Scope	Customer demographics, product sales, and recorded advertising conversions

Data publication note

The underlying course dataset is not distributed with this portfolio because the author does not hold redistribution rights. This report contains only aggregate findings and a reproducible analytical method.

Prepared as a portfolio revision of the original 2Market technical assignment.

Executive summary

The analysis connects 2,216 customer records with one-to-one advertising conversion indicators. Across six product categories, recorded sales total **\$1,345,279**, or **\$607 per customer**. Alcoholic beverages account for 50.3% of sales, making product mix the clearest commercial pattern in the data.

Spain contributes the most total sales, but it also contains approximately half of all customers. Normalized comparisons show that high market totals and high customer value are different questions. Montenegro has the highest observed sales per customer but only three customers, so that result is not decision-ready.

Twitter records 164 successful conversions, Instagram 162, and Facebook 142. Because the data contains no exposure, impression, click, spend, or campaign-timing information, these figures cannot be interpreted as conversion rates, return on investment, or causal advertising effectiveness.

Recommended decision: use the current work to identify segments for further testing, not to reallocate advertising budget. Add campaign exposure and cost data, then compare channels with conversion rates and acquisition economics.

Key measures

Measure	Result
Customers	2,216
Valid ages	2,213
Average age at 2025	55.1
Total sales	\$1,345,279
Sales per customer	\$607
Customers with any recorded conversion	459 (20.7%)

Business questions and analytical design

The original assignment combined required Excel exercises with a broader business scenario. The portfolio version separates those goals and uses a smaller set of decision-oriented questions.

Questions

- Which customer groups generate the most sales?
- Which products and countries contribute the most sales?
- How do recorded successful conversions vary across social platforms and customer segments?

Data model

Both supplied files contain 2,216 unique customer IDs, and their ID sets match. The observed relationship is one-to-one, so the current inner and left joins retain the same customers. This relationship must be revalidated whenever the data is refreshed.

Table	Grain	Primary content
Marketing	One row per customer	Demographics, product spending, activity, country
Advertising	One row per customer	Five binary successful-conversion indicators

Cleaning decisions

Three records contain implausible birth years: 1894, 1900, and 1901. They are retained in the raw data and excluded only from age-based calculations. Age is calculated with a fixed 2025 reference year instead of a volatile TODAY formula. This produces a reproducible average age of 55.1 across 2,213 valid records.

Terminology

Product values are described as sales or customer spending. They are not called profit because no costs or margins are available. Advertising fields are described as recorded successful conversions, not effectiveness, because the denominators and campaign costs are missing.

Original Tableau dashboard

This dashboard was created in Tableau for the original assignment. The preview below was recovered from the packaged workbook because the Tableau product licence is no longer active.

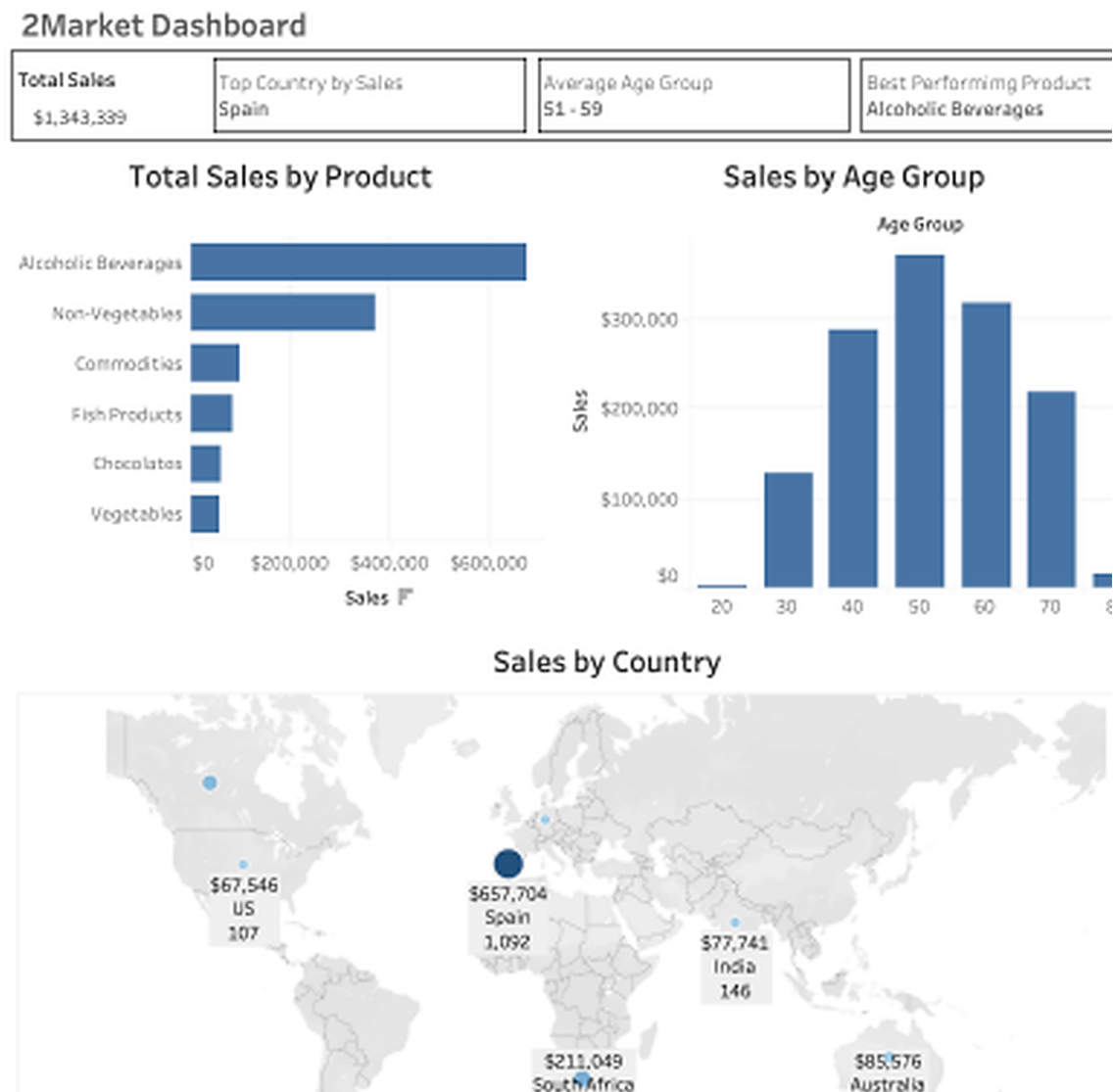
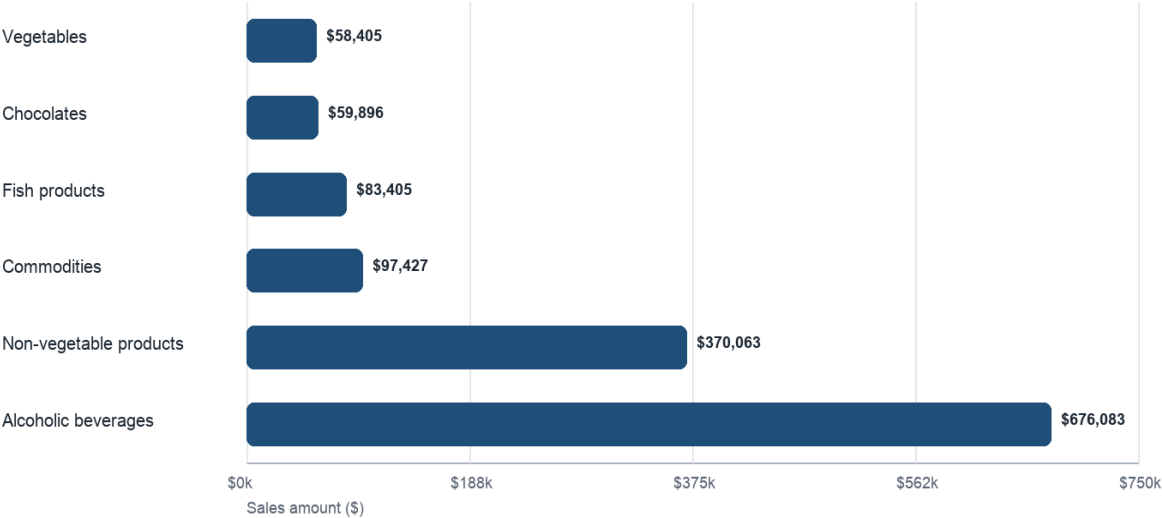


Figure 1. Original Tableau dashboard. The screenshot demonstrates the original KPI tiles, product-sales view, age-group view, country map, and filtering design. It contains aggregate results and no row-level customer information.

The screenshot is retained as evidence of the original Tableau implementation. It does not incorporate the later portfolio corrections to the age calculation, advertising-platform logic, sample-size interpretation, or normalized country comparisons. Those corrections are documented separately in the repository dashboard specification.

Product performance

Sales by product category

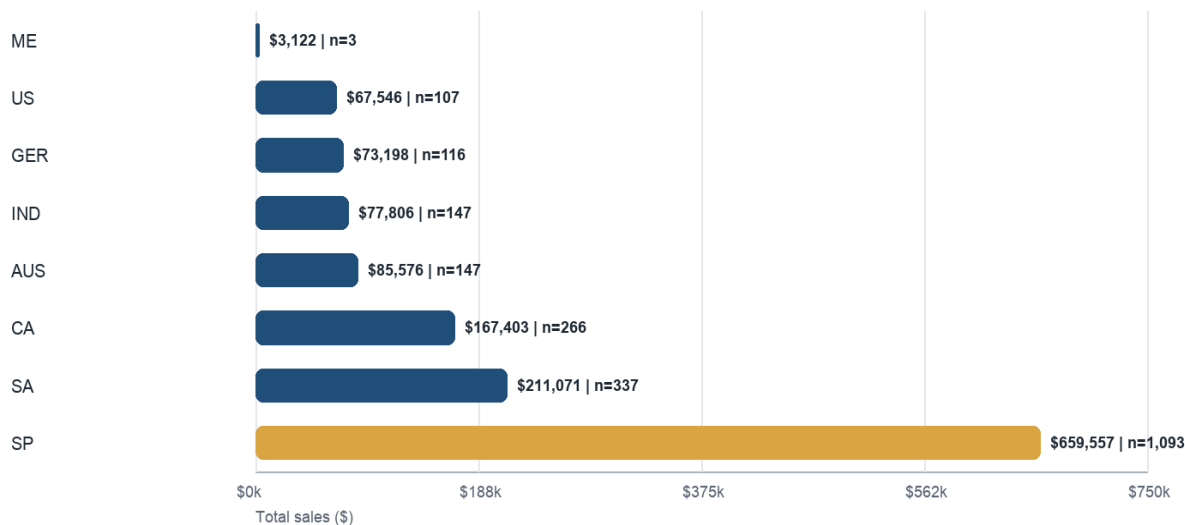


Product	Sales	Share
Alcoholic beverages	\$676,083	50.3%
Non-vegetable products	\$370,063	27.5%
Commodities	\$97,427	7.2%
Fish products	\$83,405	6.2%
Chocolates	\$59,896	4.5%
Vegetables	\$58,405	4.3%

Alcoholic beverages and non-vegetable products together account for 77.8% of recorded sales. This concentration is commercially important, but it does not establish profitability. A margin-based product decision requires cost of goods, discount, and fulfillment data.

Country performance

Country sales and customer counts



Country	Customers	Total sales	Sales/customer
SP	1,093	\$659,557	\$603
SA	337	\$211,071	\$626
CA	266	\$167,403	\$629
AUS	147	\$85,576	\$582
IND	147	\$77,806	\$529
GER	116	\$73,198	\$631
US	107	\$67,546	\$631
ME	3	\$3,122	\$1,041

Spain leads total sales because it combines substantial customer spending with the largest customer base. The United States, Germany, Canada, and South Africa each show slightly higher sales per customer than Spain. Montenegro is not comparable: its apparent \$1,041 per customer is based on only three people.

Customer age and income

After excluding three invalid birth years, the average customer age at the fixed 2025 reference year is **55.1**. The result describes this customer file; it does not prove that the customer base represents the general population.

Required income-band question

There are 41 customers with annual income from \$90,000 through \$100,000. Their average age is 52.76. Within this narrow band, the original chart reports R-squared of approximately 0.01, indicating almost no linear relationship between age and income.

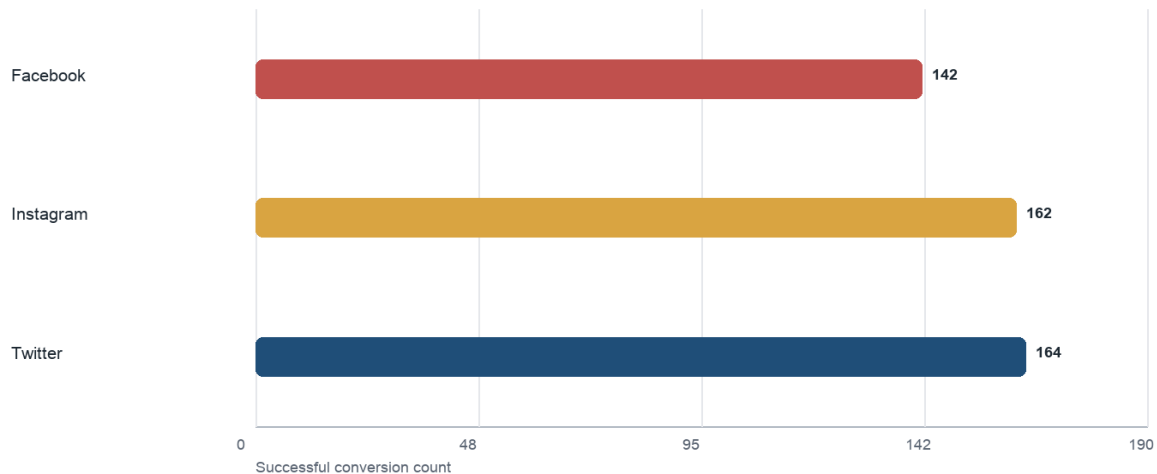
Marital-status context

Status	Customers	Average age	Sales/customer
Married	857	54.4	\$591
Together	573	56.1	\$609
Single	471	52.3	\$613
Divorced	232	57.4	\$611
Widow	76	65.2	\$728
Alone	3	51.0	\$257
Absurd	2	49.0	\$1,192
YOLO	2	51.0	\$424

The unusual labels Absurd, Alone, and YOLO contain two or three customers each. Their averages are unstable and should be reported as data-quality or category-definition issues, not as market segments.

Advertising conversions

Recorded successful conversions by social platform



Across social platforms, Twitter records 164 successful conversions, Instagram 162, and Facebook 142. The difference between Twitter and Instagram is only two conversions. A claim that Twitter is decisively superior would overstate the evidence.

Why the original platform calculation changes

A sequential Tableau IF/ELSEIF calculation assigns a multi-platform customer only to the first matching channel. The corrected design pivots the three platform columns into rows and counts each successful platform indicator independently. This preserves legitimate multi-platform records.

What cannot be calculated

Metric	Reason unavailable
Conversion rate	No exposed audience or click denominator
Cost per acquisition	No channel spend
Return on ad spend	No attributable revenue or spend
Incremental impact	No control group or experiment
Trend over time	No campaign conversion date

Use these counts to describe recorded outcomes and design follow-up tests. Do not use them alone to reallocate media budgets.

Recommendations

1. Protect the strongest categories

Maintain availability and investigate margin for alcoholic beverages and non-vegetable products. Sales concentration makes these categories important, but margin data is required before increasing investment.

2. Separate market size from customer value

Use total sales, customer count, and sales per customer together. Spain is the largest market in the data, while several smaller countries show slightly higher sales per customer.

3. Test advertising rather than declaring a winner

Run controlled, country-level tests between Twitter and Instagram. Capture exposure, clicks, spend, and conversion date so performance can be compared fairly.

4. Do not target tiny segments

Avoid recommendations based on Montenegro or the smallest marital-status labels until sample sizes are adequate and category definitions are cleaned.

5. Extend the customer analysis

Add recency-frequency-monetary segmentation and transaction dates. This would distinguish recent high-value customers from historically high spenders.

Decision framework

Decision	Current evidence	Additional data needed
Product investment	Sales mix	Margins, stock-outs, discounts
Country investment	Sales and customer count	Market costs, capacity, acquisition cost
Channel budget	Conversion counts only	Exposure, spend, clicks, timing
Customer targeting	Demographics and spending	Consent, transaction dates, retention

Limitations and portfolio improvements

The revised case study corrects the principal weaknesses in the original report while preserving the useful analytical work.

Corrections made

- Replaced profit language with sales because cost data is unavailable.
- Replaced volatile age formulas with a fixed reference year.
- Retained invalid birth-year rows in the source while excluding them from age metrics.
- Separated total country sales from sales per customer.
- Reframed platform effectiveness as recorded successful conversion counts.
- Corrected the Tableau design for customers with multiple platform conversions.
- Added sample-size warnings for small countries and marital-status groups.
- Removed unsupported claims about population representativeness and advertising causality.

Remaining limitations

- No transaction-level dates, quantities, costs, margins, or refunds.
- No advertising exposure, impressions, clicks, spend, or conversion timestamps.
- No competitor or market-size data.
- Country samples are highly unequal.
- The dataset is not distributed because publication rights are unavailable.

Conclusion

The data supports a clear descriptive story: sales are concentrated in two product categories, Spain is the largest market by total sales, and recorded social conversions are split closely between Twitter and Instagram. The analysis is useful for identifying questions and test priorities, but richer commercial and campaign data is required before making profit or advertising-allocation decisions.